

John A. Scarrow

Coeur d'Alene, ID | JohnA@Scarrows.com | (425) 802-6231
linkedin.com/in/JohnAScarrow | github.com/JohnScarrow

SUMMARY

CS student at North Idaho College with two years of C++ and Python coursework, a one-year full-stack JavaScript program (Code Fellows), and a track record of shipping real systems — from a CUDA-accelerated ML game AI compiled to WebAssembly, to a YOLOv11 wildlife detector deployed on a Raspberry Pi 5. Strong foundation in systems programming, machine learning inference, and full-stack web development.

TECHNICAL SKILLS

- **Languages:** C++17, Python, TypeScript, JavaScript, x86-64 Assembly, CUDA, SQL
- **ML / CV:** YOLOv11, OpenCV, ONNX Runtime, WebAssembly (Emscripten), INT8 quantization
- **Web / Backend:** React, Fastify, Express.js, Node.js, REST APIs, Prisma ORM, PostgreSQL, MongoDB
- **Cloud / DevOps:** Docker, AWS, GitHub Actions, Vercel, Railway
- **Tools:** SFML, NASM, Git

SELECTED PROJECTS

Battleship-ML (C++17 · CUDA · WebAssembly)

- Built a Battleship AI that blends four probabilistic signals per shot: a log-learned global heatmap, a live ship-placement coverage map, Monte Carlo sampling, and tactical bonuses (adjacency, parity, fit-score)
- Implemented a CUDA kernel for GPU-accelerated Monte Carlo sampling using cuRAND and shared-memory atomics; falls back to CPU automatically
- Native CLI tuner supports multi-threaded grid-search over AIWeights parameters and an online-learning mode
- Compiled the full C++ engine to WebAssembly via Emscripten for a live browser demo with a real-time probability heatmap

AutoDetect-Wildlife (Python · YOLOv11 · ONNX · Raspberry Pi)

- Built a real-time wildlife detection system with two targets: YOLOv11s on a laptop (60–80 FPS) and a custom INT8-quantized ONNX model on a Raspberry Pi 5 (~3 FPS, no PyTorch required)
- Wrote a training pipeline that downloads and merges Roboflow and LILA.science datasets, trains separate models per platform, and exports the Pi model to INT8 ONNX
- Implemented custom letterbox preprocessing and YOLO NMS postprocessing on the Pi path; supports headless MJPEG stream over HTTP

Wild West Shootout: High Noon (x86-64 Assembly)

- Built a fully playable shootout game in x86-64 Assembly using NASM — no C runtime, no standard library
- Managed the call stack, register allocation, and all control flow by hand using direct Linux system calls
- Demonstrated practical understanding of CPU architecture, memory layout, and low-level I/O

Bounty of Juan (C++17 · SFML)

- 4-person team project: 2D top-down survivor shooter with infinite tiling world, wave-based enemy spawning, auto-targeting projectiles, and live HUD
- Owned health system, results screen, ESC pause overlay, and enemy contact-damage windup animation
- Used a Character base class with polymorphic takeDamage / isDead shared between player and enemy types

Project Insight Hub (TypeScript · React · Fastify · PostgreSQL)

- Rebuilt a non-functional full-stack codebase into a working prototype — diagnosed broken state, rewired backend auth, and got the full JWT + RBAC flow operational
- Three-tier role system (Admin, Editor, Viewer), hierarchical task tracking, time logging, cost monitoring, and audit logging
- Stack: React, Vite, ShadCN UI, Tailwind, TanStack Query; Fastify, Prisma, PostgreSQL, bcrypt

EXPERIENCE

Freelance Developer — Coeur d'Alene, ID

March 2020 – Present

- Designed and maintained full-stack web applications for clients
- Implemented backend features, debugging, and iterative improvements
- Collaborated with clients to deliver stable, maintainable solutions

Snoqualmie Tribe Market — Snoqualmie, WA

Supervisor, POS Support | March 2018 – May 2020

- Troubleshoot POS systems and resolved technical issues under operational pressure
- Managed staff scheduling and ensured day-to-day reliability

Relectromix LLC — Bellevue, WA

Technician | March 2016 – April 2017

- Repaired PC/Mac systems and configured Windows Server environments
- Performed diagnostics and customer-facing technical support

EDUCATION

North Idaho College — Current Student

C++, Python, Robotics, and Computer Science coursework (2024 – Present)

Code Fellows, Seattle — Full-Stack JavaScript Development Program (2017–2018)

Bellevue College — Full-Stack JavaScript Coursework (2015–2017)

Eastlake High School — Graduate (2011)

REFERENCES

Available upon request.

Academic reference: **Professor Ekaterina Miller**, North Idaho College.